

The Poisson Distribution



Conditions for a Poisson distribution

- 1 State the key conditions required for a random variable to be modeled by a Poisson distribution.
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Computing Poisson probabilities

- 2 A call center receives an average of 4 calls per minute, modeled by a Poisson distribution with $\lambda = 4$. Find $P(X = 6)$, using $P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$, to three decimal places.
 - 3 Using the same call center ($\lambda = 4$ calls per minute), find the probability of receiving at most 2 calls in a given minute.
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Mean and variance of a Poisson distribution

- 4 State the mean and variance of a Poisson distribution with parameter λ , and explain what is distinctive about this relationship compared to other common distributions.
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Sums of independent Poisson variables

- 5 Two independent Poisson processes have rates $\lambda_1 = 3$ and $\lambda_2 = 5$ events per hour. If $X_1 \sim \text{Po}(3)$ and $X_2 \sim \text{Po}(5)$, state the distribution of $X_1 + X_2$, and find its mean and variance.
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Poisson as an approximation to the binomial

- 6 Explain the conditions under which a binomial distribution $X \sim B(n, p)$ can be well-approximated by a Poisson distribution, and state the appropriate value of λ to use.
 - 7 A factory produces 2000 items, each independently defective with probability 0.002. Using the Poisson approximation, find the probability that exactly 3 items are defective, to three decimal places.
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Using cumulative Poisson probabilities

- 8 A shop receives customers at an average rate of $\lambda = 5$ per 10-minute interval. Find the probability of receiving more than 2 customers in a given 10-minute interval, given $P(X = 0) \approx 0.0067$, $P(X = 1) \approx 0.0337$, and $P(X = 2) \approx 0.0842$.

Finding an unknown parameter from a given probability

- 9 A random variable $X \sim \text{Po}(\lambda)$ satisfies $P(X = 0) = 0.1353$. Find λ , to the nearest whole number, using $P(X = 0) = e^{-\lambda}$.
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Variance-based reasoning with the Poisson distribution

- 10 A random variable $X \sim \text{Po}(\lambda)$ has standard deviation 3. Find λ , and then find $P(X = \lambda)$, to three decimal places.
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Combining Poisson reasoning with real-world decisions

- 11 A website server crashes on average $\lambda = 0.5$ times per week. The IT team wants to know the probability of at least one crash in a given week, given $P(X = 0) \approx 0.6065$. Should the team be concerned if their target is to keep this probability below 30%?
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Combining a Poisson approximation with conditional probability

- 12 Machine faults occur at an average rate of $\lambda = 1.5$ per day. Find the probability that exactly one fault occurs today, given that at least one fault occurs today, to three decimal places, using $P(X = 0) \approx 0.2231$ and $P(X = 1) \approx 0.3347$.
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Combining Poisson processes over different time intervals

- 13 A radioactive source emits particles at a rate of $\lambda = 2$ per second, modeled by a Poisson distribution. Find $P(X = 3)$ for a 3-second interval, to three decimal places, using the fact that the parameter scales with the length of the interval.
- 14 Explain why the Poisson distribution is not an appropriate model for the number of heads obtained when flipping a fair coin 10 times, referencing the conditions required for a Poisson model.
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Comparing two independent Poisson processes

- 15** Two independent Poisson processes have rates $\lambda_A = 3$ and $\lambda_B = 2$ per hour. Find the probability that, in a given hour, process A produces exactly 2 events and process B produces exactly 1 event, to three decimal places, given $P(X_A = 2) \approx 0.2240$ and $P(X_B = 1) \approx 0.2707$.